

Chapter 2:

Research Methods

Today's agenda

- Review basic research terms
- Why do we measure behavior?
- Examine fly behavior (video)

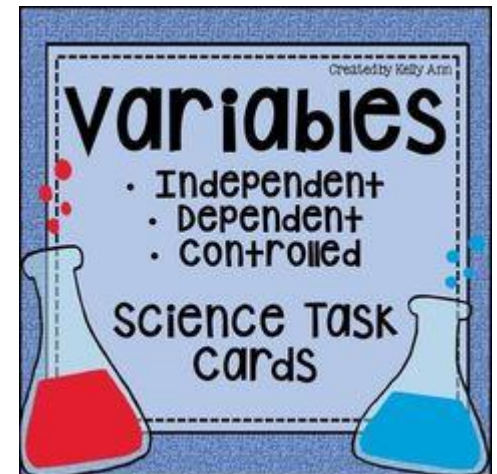
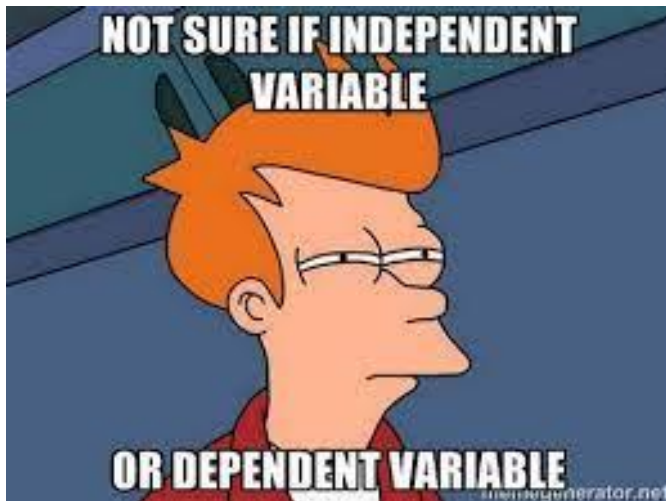
Behavioral Research

- Begin by developing a hypothesis or by deciding on a specific area of interest
- Employ a research method
 - naturalistic observation
 - case studies
 - control group designs
 - single-subject designs
- Collect behavioral data



Variables

- Characteristics of a person, place, or thing that can change (vary) over time or from one situation to another.
 - Examples: height, weight, marital status, age, hair color



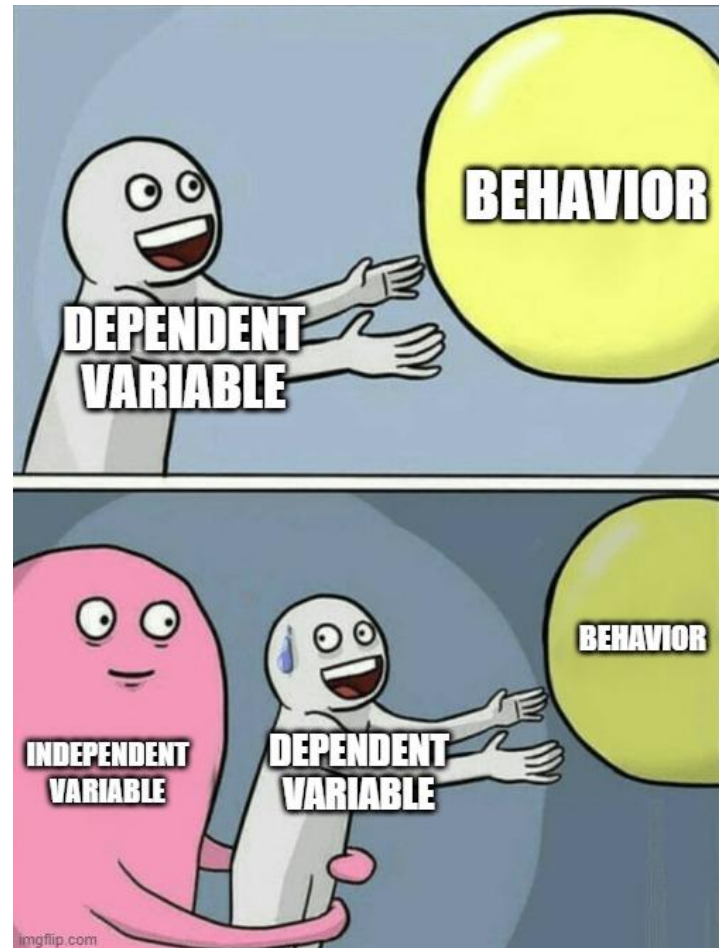
Independent Variable

- Systematically varies across different conditions in an experiment.
- They are manipulated by the experimenter.
- Example:
 - The direction of the looming stimulus for the fly
 - Room temperature
 - Drug treatment



Dependent Variable

- The aspect of an experiment that is allowed to vary freely to see if it is affected by changes in the independent variable.
- They are measured by the experimenter.



What kinds of behaviors can a fly perform?

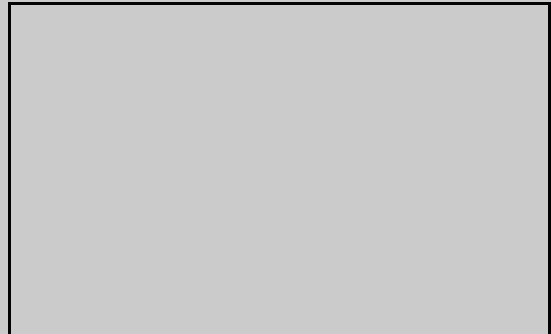
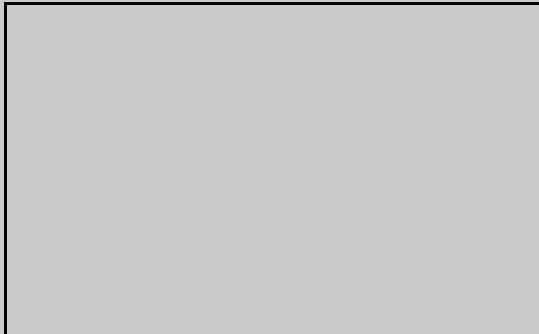
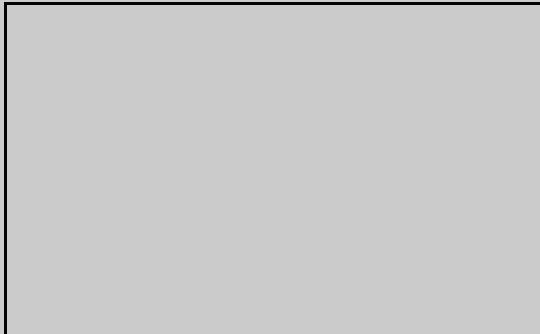


Fly behavior

Side view 1 (xz)

Side view 2 (yz)

Top view (xy)



Fly Tournament



What can a fly do?

What kind of behaviors can you measure?

So what can we measure

- Latency to first contact
- Duration of interaction
- Offensive behavior, defensive behavior
- Distance from food

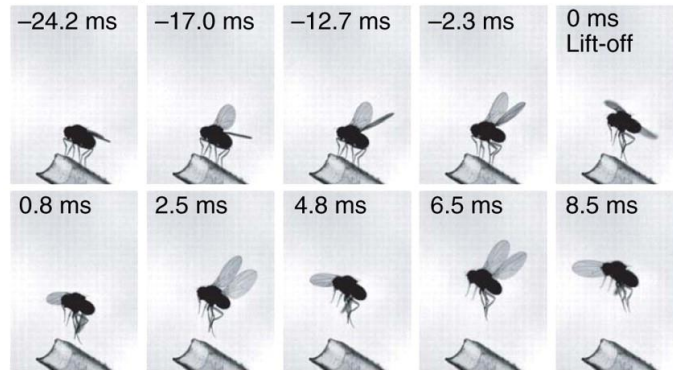
Fly take off and decision making?



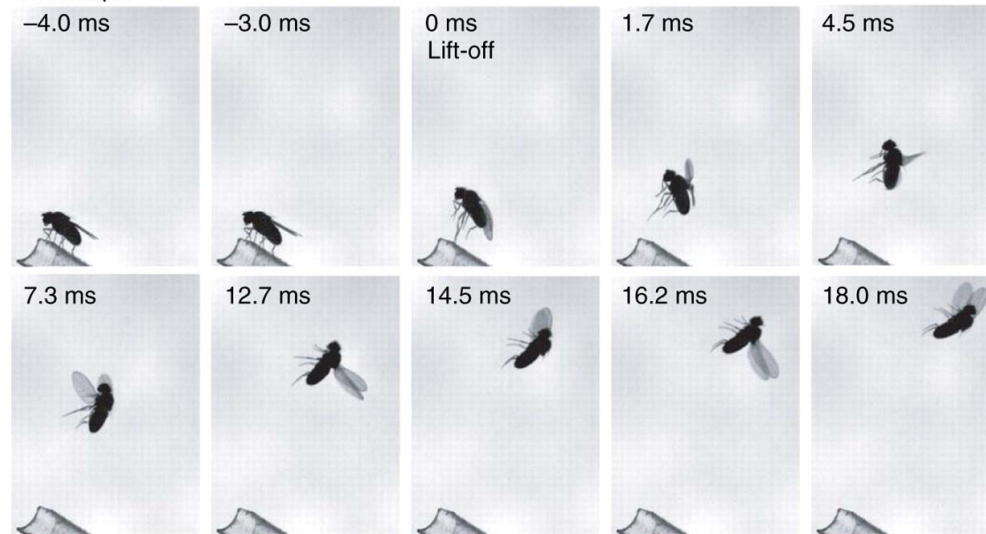


Video sequences for (A) voluntary and (B) escape take-offs

A Voluntary

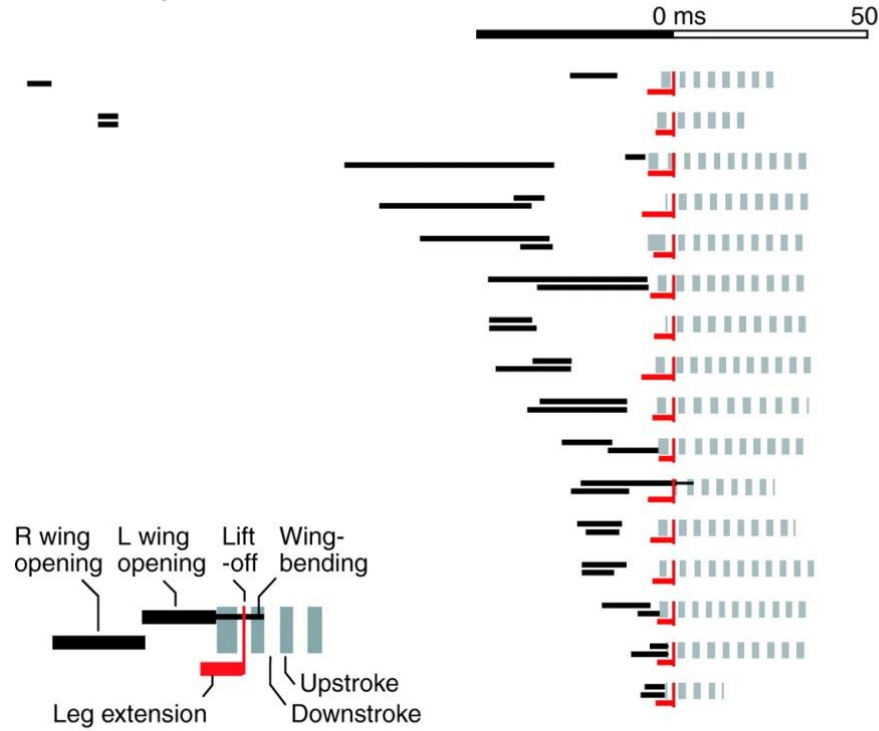


B Escape

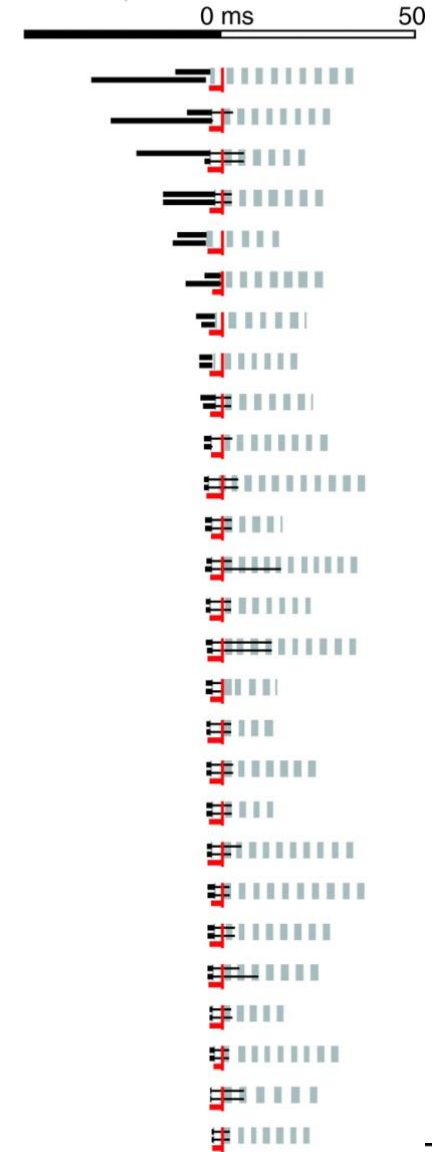


Timelines of leg and wing events during (A) voluntary (N=16) and

A Voluntary



B Escape



Fly Behavior-Independent variable



Functional Relationship

- The relationship between changes in an independent variable and changes in a dependent variable.
- Cause-and-effect relationship.

State-dependent decoupling of sensory and motor circuits underlies behavioral flexibility in *Drosophila*

Jan M. Ache ¹, Shigehiro Namiki^{1,2}, Allen Lee^{1,3}, Kristin Branson ¹ and Gwyneth M. Card ^{1*}

An approaching predator and self-motion toward an object can generate similar looming patterns on the retina, but these situations demand different rapid responses. How central circuits flexibly process visual cues to activate appropriate, fast motor pathways remains unclear. Here we identify two descending neuron (DN) types that control landing and contribute to visuo-motor flexibility in *Drosophila*. For each, silencing impairs visually evoked landing, activation drives landing, and spike rate determines leg extension amplitude. Critically, visual responses of both DNs are severely attenuated during non-flight periods, effectively decoupling visual stimuli from the landing motor pathway when landing is inappropriate. The flight-dependence mechanism differs between DN types. Octopamine exposure mimics flight effects in one, whereas the other probably receives neuronal feedback from flight motor circuits. Thus, this sensorimotor flexibility arises from distinct mechanisms for gating action-specific descending pathways, such that sensory and motor networks are coupled or decoupled according to the behavioral state.



VT50660-GAL4
/ UAS-trpA1 @ RT



VT50660-GAL4
/ UAS-trpA1 @ 30°C

Stimulus

- any event that can potentially influence behavior.
- Examples:
 - Events both outside the skin and inside the skin can influence our behavior.



Response

- A particular instance of a behavior.
- The response of one organism can act as a stimulus that influences the response of another organism.
- Example:
 - When one fly fights the other



- What kind of stimuli can be found in a car?
- Safety features?



Mazda CX-9 safety features

- Blind Spot Monitoring with Rear Cross Traffic Alert⁹
Smart City Brake Support with collision warning³³
Advanced Smart City Brake Support with pedestrian detection³⁴
Lane-keep Assist¹³
Lane Departure Warning¹²
Mazda Radar Cruise Control with Stop & Go¹⁴

Overt Behavior

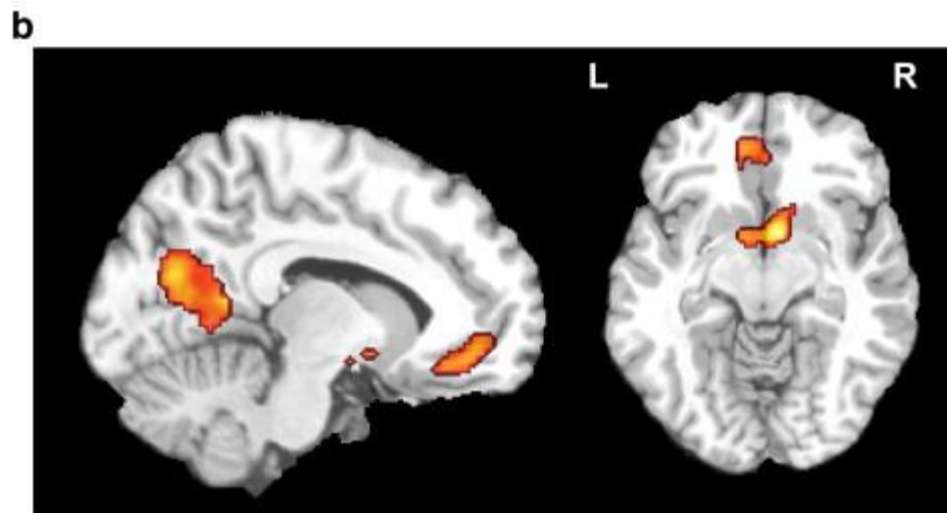
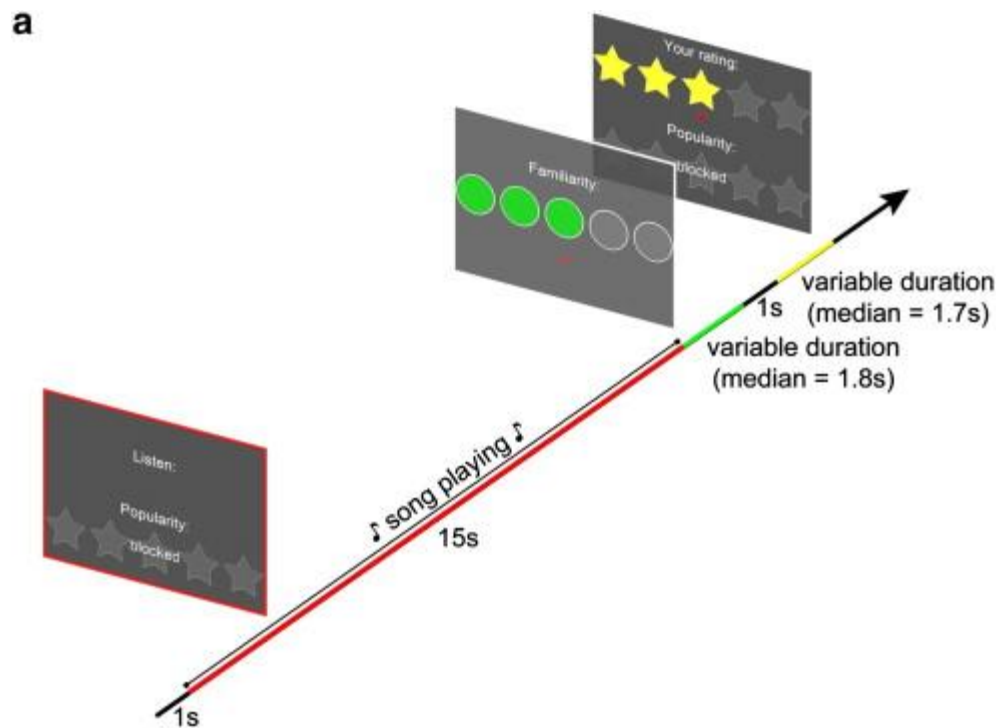
- Is behavior that has the potential for being directly observed by an individual other than the one performing the behavior.
- Examples:
 - response of saying hello
 - Escape direction



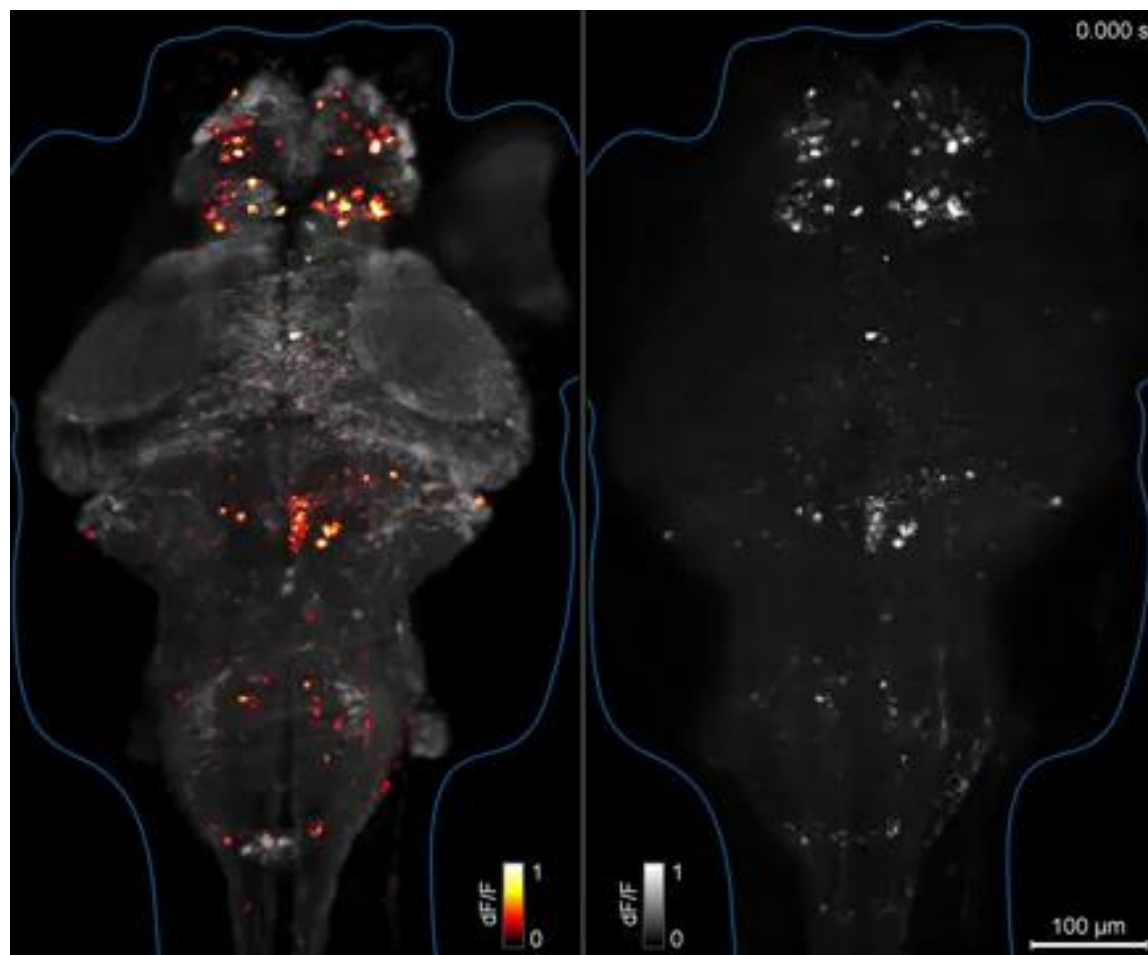
Covert Behavior

- behavior that can be perceived only by the person performing the behavior.
- Examples:
 - Feeling anxiety
 - Fear for the fly





Gregory S. Berns, Sara E. Moore. **A neural predictor of cultural popularity.** *Journal of Consumer Psychology*, 2011



Talking to Strangers



Malcolm
Gladwell

#1 NEW YORK TIMES bestselling author of *OUTLIERS*
and host of the podcast *REVISIONIST HISTORY*

- How do you know if someone is lying?

Appetitive Stimulus

- An event that an organism will seek out.
- Examples:
 - Food when we are hungry
 - Water when we are thirsty
- These events are not always “pleasant”.
- Examples:
 - Smokers seek out cigarettes even though they feel smoking is disgusting.

Aversive Stimulus

- An event that an organism will avoid.
- Examples:
 - Extreme heat
 - Electric shock
- These events are not always “unpleasant”.





Contents lists available at [ScienceDirect](#)

Psychiatry Research

journal homepage: www.elsevier.com/locate/psychres



Self-inflicted pain out of boredom

Chantal Nederkoorn^{a,*}, Linda Vancleef^a, Alexandra Wilkenhöner^a, Laurence Claes^b,
Remco C. Havermans^a

^a Department of Clinical Psychological Science, Maastricht University, Maastricht, The Netherlands

^b Department of Clinical Psychology, KU Leuven, Leuven, Belgium

Motivating Operation

- A procedure that affects the appetitiveness or aversiveness of a stimulus.
- Appetitiveness or aversiveness of an event depends on a particular state or condition.

- Establishing operation
 - Increases appetitiveness or aversiveness of an event
- Abolishing operation
 - Decreases appetitiveness or aversiveness of an event

Deprivation

- The prolonged absence of an event that tends to increase the appetitiveness of that event.
- Examples:
 - Going without food, television, or social contact for a long period of time, texting



EDITORIAL • 14 JANUARY 2019

Screen time: how much is too much?

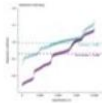
A vast analysis tackles a defining question of the digital age.



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China Tightens Limits for Young Online Gamers and Bans School Night Play

The restrictions reflect the government's intensifying push for companies to jettison what it says are unhealthy cultural influences.



Satiation

- The prolonged exposure to an event, which tends to decrease the appetitiveness of that event.
- Examples:
 - Eating a large meal
 - Listening to the same song/music too many times



Other ways our behavior is influenced:

- Contiguity
 - Temporal



Other ways our behavior is influenced:

- Contiguity
 - Temporal
 - Spatial



Other ways our behavior is influenced:

- Contiguity
 - Temporal
 - Spatial
- Contingency



EVERYDAY FAVORITES

ALL SIZES UNDER 200 CALORIES

	TALL 12 FL OZ	GRANDE 16 FL OZ	VENTI 20 FL OZ
Pike Place Roast	1.75 5 cal	1.95 5 cal	2.10 5 cal
Brewed Hot Tea	1.75 0 cal	2.15 0 cal	2.15 0 cal
Caffè Americano	2.10 10 cal	2.40 15 cal	2.80 25 cal
Cappuccino	3.20 90 cal	3.75 120 cal	4.05 150 cal
Skinny Vanilla Latte	3.50 90 cal	4.05 130 cal	4.35 160 cal
Skinny Caffè Mocha	3.50 90 cal	4.05 130 cal	4.35 160 cal
Skinny Cinnamon Dolce Latte	3.90 90 cal	4.45 130 cal	4.75 160 cal
	SOLO 1 FL OZ	DOPPIO 2 FL OZ	
Espresso	1.75 5 cal	2.00 10 cal	
	TALL 12 FL OZ	GRANDE 16 FL OZ	VENTI 24 FL OZ
Iced Brewed Coffee	2.10 60 cal	2.40 90 cal	2.80 130 cal
Iced Vanilla Brewed Coffee	2.10 60 cal	2.40 90 cal	2.80 130 cal
Shaken Iced Tea Available in Black, Green or Passion*	1.85 60 cal	2.10 80 cal	2.40 130 cal
Shaken Iced			

Whit
Cinn

Iced

Serving Up

McDonald's plans to show the calories for all its menu items. Here's how the Big Mac compares to menu items from other food chains:



Big Mac
550
calories



**TACO
BELL**

Cantina Burrito
760
calories

**BURGER
KING**

Whopper*
600
calories

STARBUCKS

Venti Java Chip
Frappuccino†
450
calories

Panera

Cinnamon
crunch bagel
420
calories

*With cheese, no mayo †Soy milk, no whipped cream
Source: the companies

The Wall Street Journal



Measurements

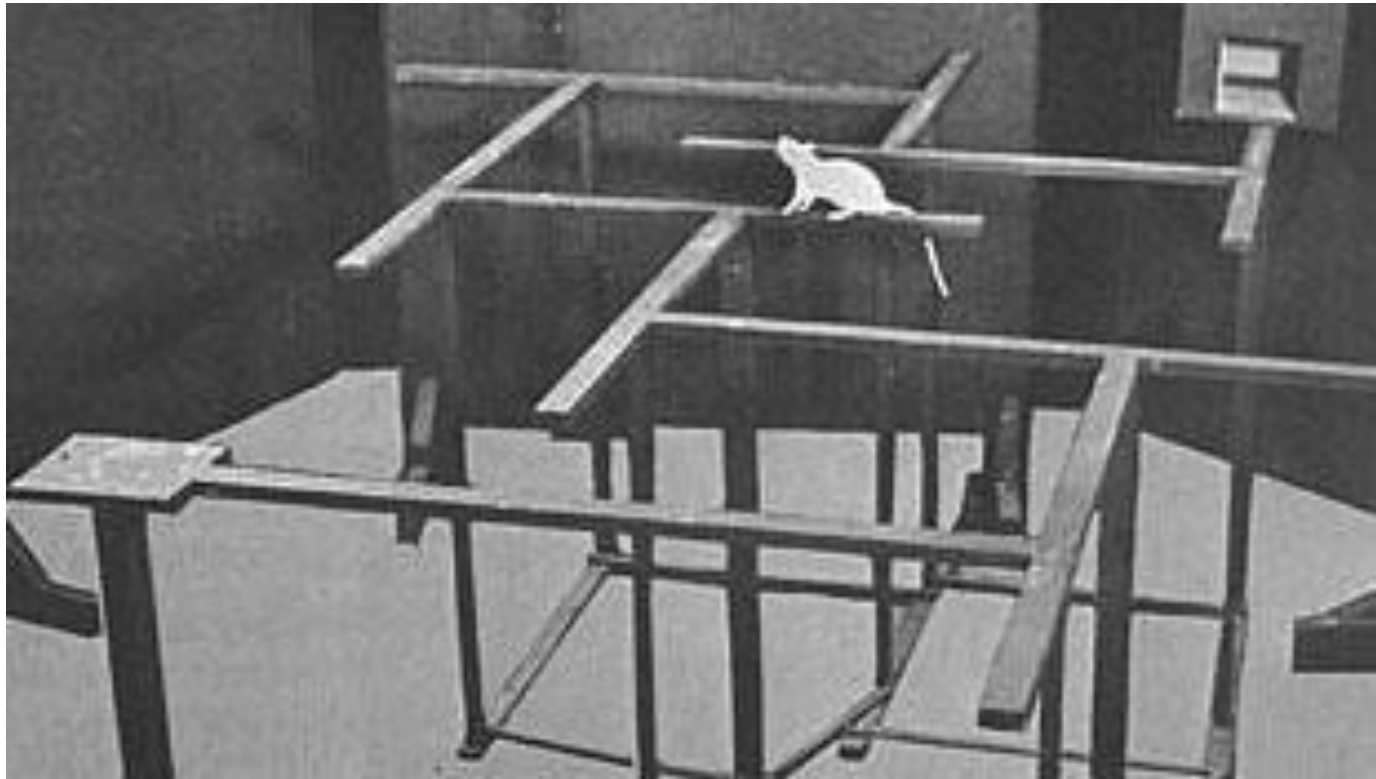
- Where we will start Thursday

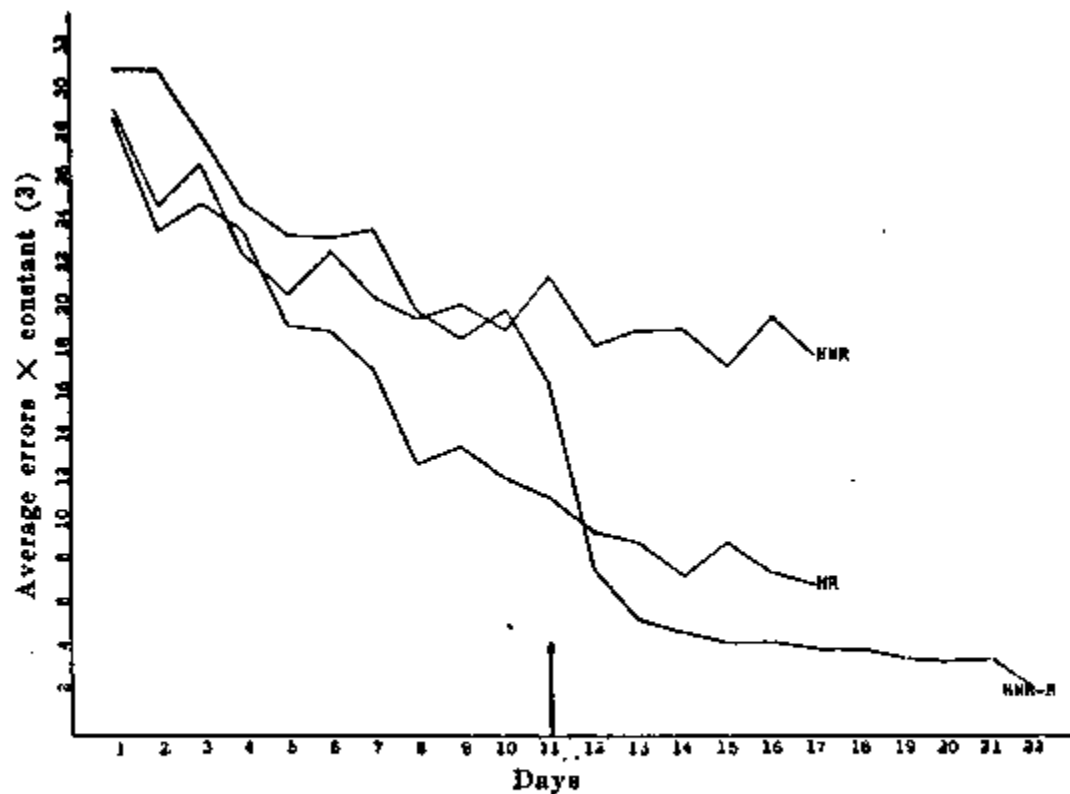
Measurements

- Rate of Response
- Intensity
- Duration
- Latency
- Number of errors
- Speed



Number of Errors





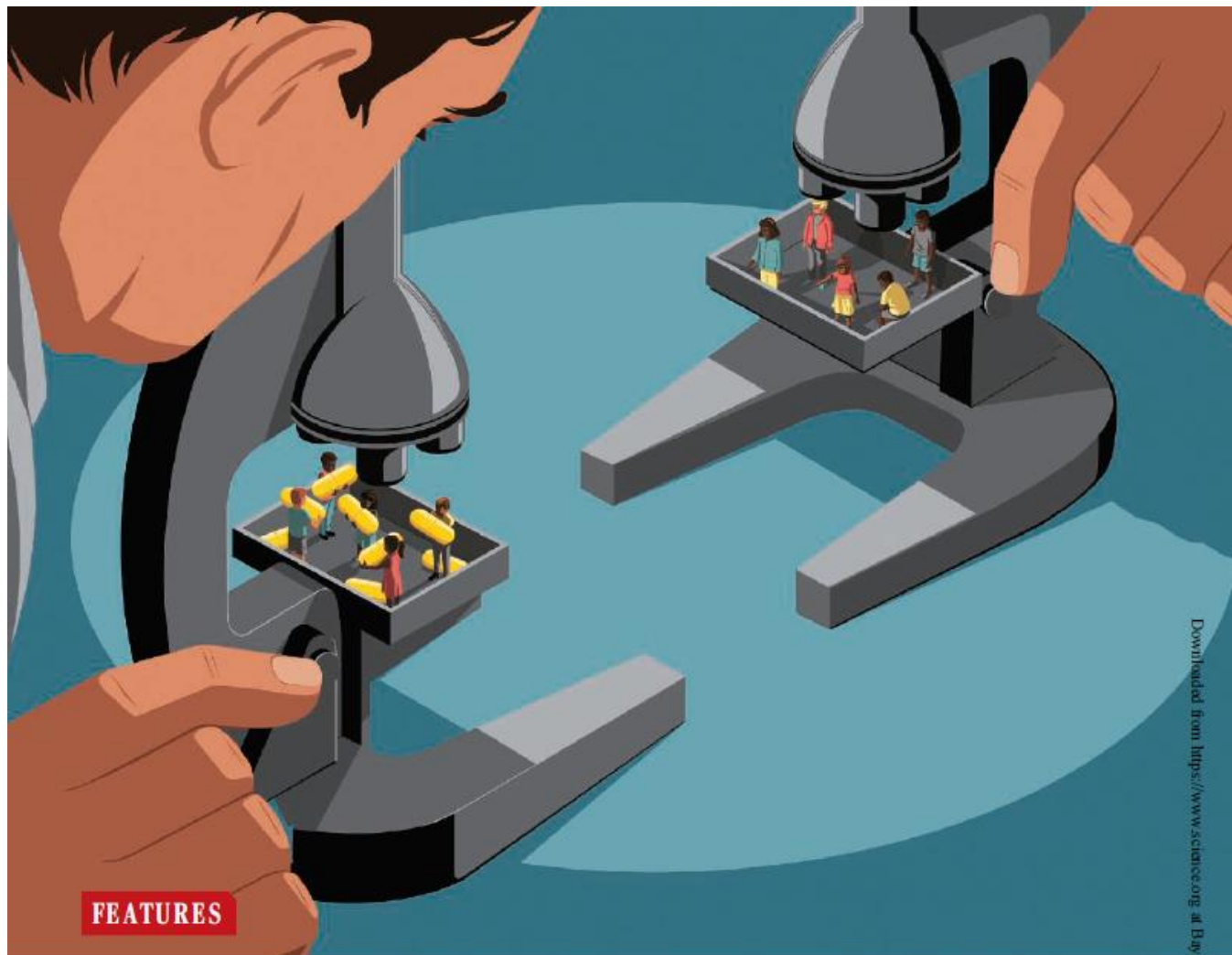
Error curves for HR, HNR, and HNR-R

FIG. 6

(From E. C. Tolman and C. H. Honzik, Introduction and removal of reward, and maze performance in rats. *Univ. Calif. Publ. Psychol.*, 1930, 4, No. 19, p. 267.)

Experimental Research

- Experiments to discover cause-and-effect relationships between environmental events and behavior.
- One or more independent variables are systematically varied to determine their effect on a dependent variable.
- The two main types are control group designs and single-subject designs.



FEATURES

FAILURE TO PROTECT?

A study of asthmatic children, most of them Black, shows how a common clinical trial design can expose vulnerable participants to serious risks

Control Group Designs

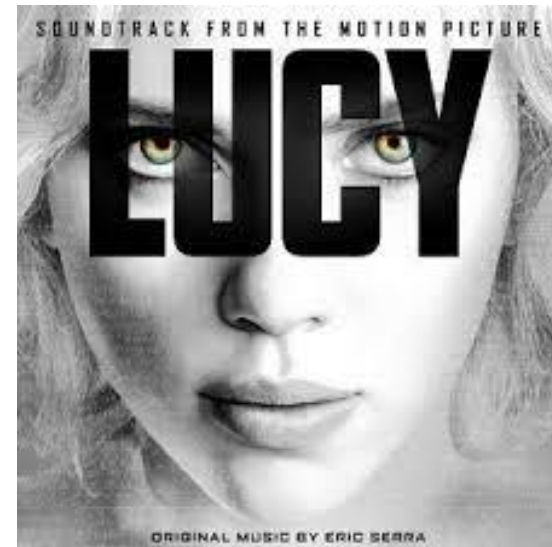
- Individuals are randomly assigned to either an experimental group or a control group
- Individuals assigned to the experimental group are exposed to a certain manipulation or treatment
- Those assigned to the control group are not exposed to the manipulation

Factorial Design

- Control group designs that examine the interaction of more than one independent variable.
- Example:
 - We might rerun the experiment (free food versus no food) with groups of old rats, middle-aged rats, and young rats.
 - This is a 2×3 *factorial design*.
 - Interaction effects give us a much finer understanding of the variables.

Memory Enhancing Drug Trial

Young	Middle Age	Older Adult
No drug		
Memory Drug		



Mask type

	Young Adult	Older Adult	Other conditions?
None			
Cloth			
Neck gaiter			
Surgical			
Any groups missing?			

Benefits of Control Group Designs

- You can assess the general effects of certain variables.
- You can determine cause-and-effects.
- You can be fairly confident that differences between groups in performance are the result of differences in the independent variable.
- Various characteristics of the subjects in each group are likely to be evenly distributed.

Limitations of Control Group Designs

- They require a large number of subjects.
- They are not well suited for investigating the effect of a certain treatment on a particular individual.
- They focus on the *average* performance of all subjects in each group.
- Results are often analyzed and interpreted only at the end of the experiment.
 - What about life-threatening diseases?

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Early end for HRT clinical trial

Combination of estrogen and progestin proves more risky than healthful, triggering halt of first definitive, large, national study of hormone therapy

JULY 18, 2002 · BY BARBARA BERG



A Women's Health Initiative study of a combination of the hormones estrogen and progestin, prescribed under the brand Prempro, was stopped early because of results

25 JUNE 2020

COMMENT

Dexamethasone, an unexpected drug, shows promise in treating Covid-19

By GlobalData Healthcare

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RECOMMENDED COMPANIES



Jeeva Informatics Solutions

Jeeva Informatics Solutions (Jeeva) is a seed-stage start-up on a...



Experic

Experic is a contract manufacturing organization (CMO) and pharmaceutical supply...



Malvern Panalytical

Malvern Panalytical offers characterisation tools for molecular analysis and

FDA cautions against use of hydroxychloroquine or chloroquine for COVID-19 outside of the hospital setting or a clinical trial due to risk of heart rhythm problems

Does not affect FDA-approved uses for malaria, lupus, and rheumatoid arthritis

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July 1, 2020 Update: A summary of the FDA [review of safety issues](#) with the use of hydroxychloroquine and chloroquine to treat hospitalized patients with COVID-19 is now available. This includes reports of serious heart rhythm problems and other safety issues, including blood and lymph system disorders, kidney injuries, and liver problems and failure.

June 15, 2020 Update: Based on ongoing analysis and emerging scientific data, FDA has revoked the emergency use authorization (EUA) to use hydroxychloroquine and chloroquine to treat COVID-19 in certain hospitalized patients when a clinical trial is unavailable or participation is not feasible. We made this determination based on recent results from a large, randomized clinical trial in hospitalized patients that found these medicines showed no benefit for decreasing the likelihood of death or speeding recovery. This outcome was consistent with other new data, including those showing the suggested

Content current as of:
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Regulated Product(s)
Drugs

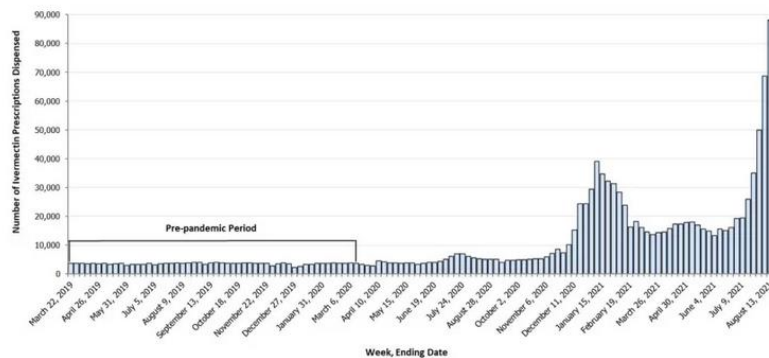
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Safety - Issues, Errors, and Problems

Health Topic(s)
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Coronavirus



CALIFORNIA

Anti-vaccine forces pushing ivermectin. It can be toxic, dangerous, officials say



The number of outpatient ivermectin prescriptions dispensed from retail pharmacies. (U.S. Centers for Disease Control and Prevention)

BY RONG-GONG LIN II | STAFF WRITER

Los Angeles Times

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☐ Recruiting and not yet recruiting studies

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Condition or disease (For example: breast cancer)

Other terms ⓘ (For example: NCT number, drug name, investigator name)

X

Country ⓘ

_____ x _____

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Single-Subject Designs

- Require only one or a few subjects to conduct an entire experiment.
- The problem of generalizability
- There are several types.



CHALLENGE

SIMPLE. DELICIOUS. EFFECTIVE



The 28-Day Keto Challenge

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The 28-Day Keto Challenge gives you an easy-to-follow plan and includes everything you need to be successful with the Keto diet - both now and in the long term.

EXPLORE **Health**

Your Ac

RELATED: [What Is Opiah's Latest Diet? A Nutritionist Explains the OMD](#)[Plan](#)

One of this year's worst diets (and biggest surprises): keto.

Yep, the high-fat, low-carb diet which has amassed a cult following due to its get-slim-quick promises, ranked second-to-last with experts noting concerns with its emphasis on fat-rich foods.

"This diet is fundamentally at odds with everything we know about long-term health," one expert [pointed out](#). They also noted it is "minimally effective" in both preventing diabetes and promoting heart health, has insufficient evidence supporting long-term weight loss, ("It hasn't been used for weight management long enough to establish a strong track record for helping people lose weight and keep it off long term," says Sass) isn't very nutritious, and is difficult to follow. "Following this eating plan long term is the most difficult aspect of this diet regime," one expert said. "People become very bored just eating fatty foods, fat and meat." Another actually described it as "the hardest of the extremely hard!"

In addition to being difficult to maintain, Sass adds that keto also has the potential to create negative health outcomes, including nutrient deficiencies, and alter the gut microbiome in ways that may increase inflammation and impact immunity and mental health.

The only category where keto fared reasonably well, was Best Fast Weight-Loss Diets, tying for third place.

Other low-ranking diets included the Dukan Diet, Whole30, Atkins, and the Raw Food diet. "The number one thing the five diets at the bottom of the list have in common is a lack of research, either about weight loss outcomes, or their impact on health," Sass points out.



I'm Lovin' it!

Peter Travers, Rolling Stone

"Funny and outrageous"
Owen Gleiberman
Entertainment Weekly

"Two thumbs up!"
David Karger

SUPER SIZE ME

A Film of Epic Portions

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Simple-Comparison Design

- Behavior
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AVERAGE GYM DAYS 0 AVERAGE GYM DAYS 7



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Simple-Comparison Design

- This design does not control for the possibility that some other event that occurred at the same time as the treatment could have caused the change in the behavior.
 - Example treating for caffeine consumption during finals week or cigarette consumption during hurricane season

Other examples

- Weight loss
 - You break your leg
 - Christmas/Thanksgiving time
 - Backpacking trip

Single-Comparison Design

- One benefit is that you just need one or a few people to run the study
- It is a great way to change our own behavior or someone we know

How can I get them to help with
house duties to achieve their goal?
What did your parents do?

Cleaning their room

- What are important considerations?
- What are you going to measure?
- Treatment?

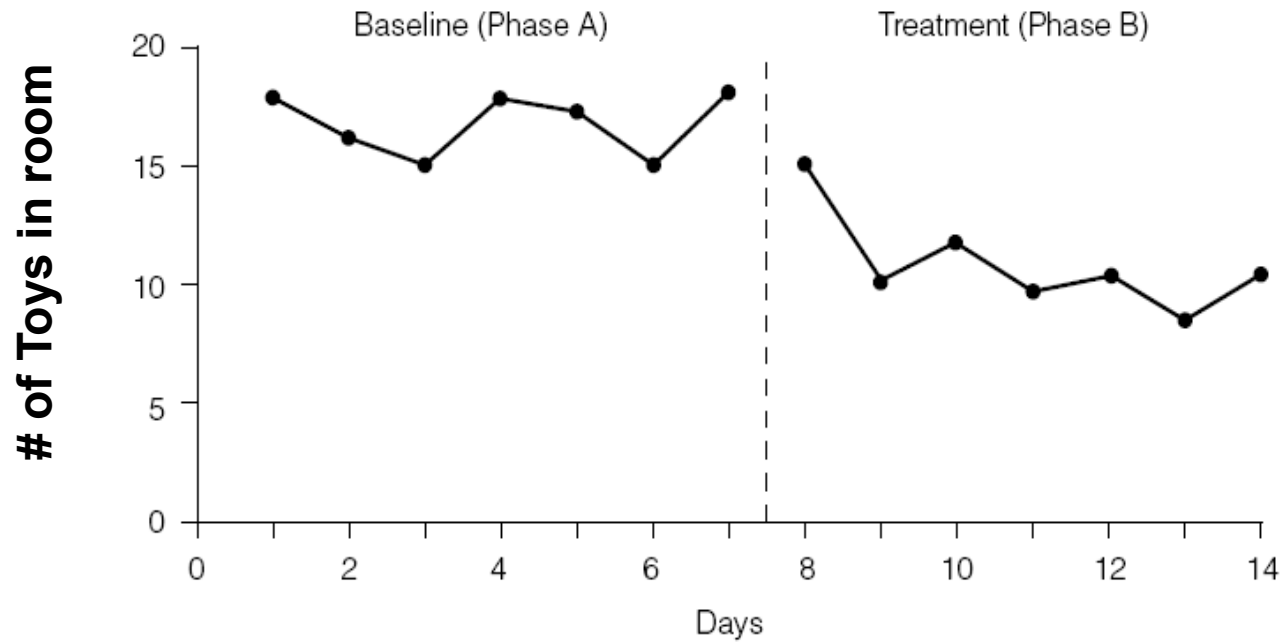
What is our experiment?

- What is our measurement?
- When do we measure it?

The reward



Pick up toys behavior



How do I know that the behavior is due to the treatment?

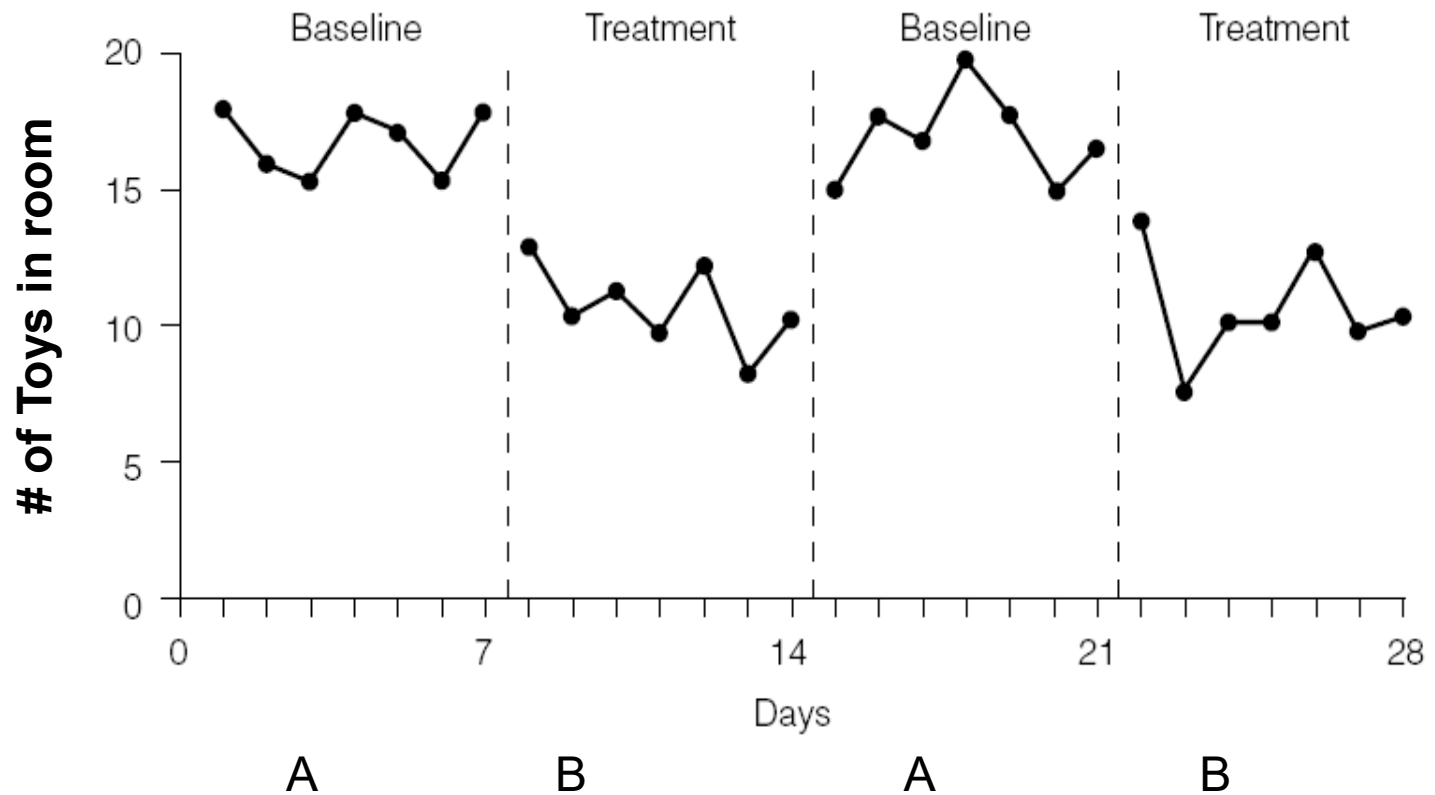
- What would you do?

Reversal Design

- Involves repeated alternations between a baseline period and a treatment period.
- If the behavior changes each time the treatment is instituted and later withdrawn, then a functional relationship has been demonstrated.

Reversal Design Results

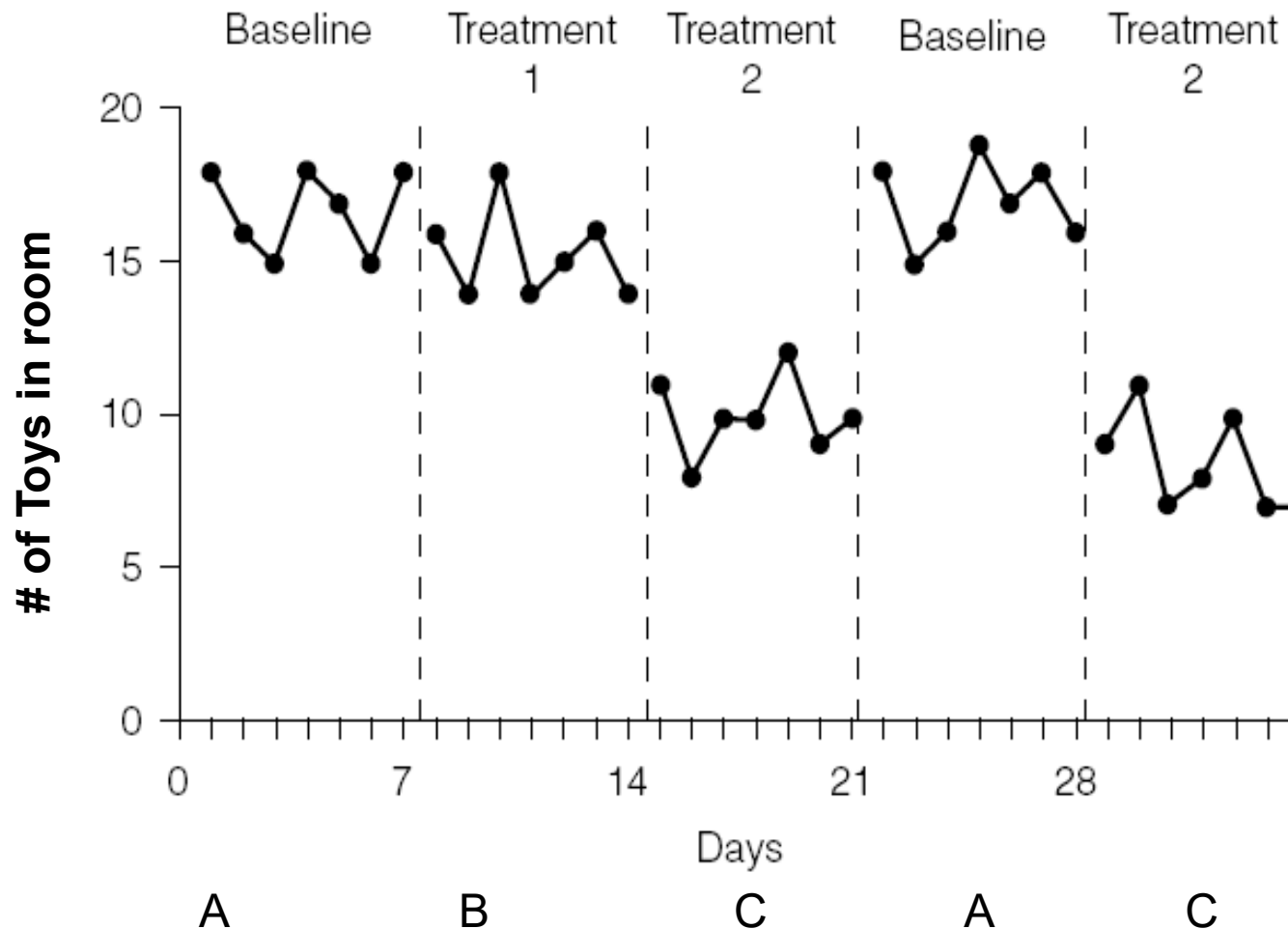
Would this happen?



Limitations of Reversal Designs

- The behavior must revert to its original baseline frequency when the treatment is withdrawn.
- Some treatments are intended to produce long-lasting effects.
- It may be ethically inappropriate to remove a treatment once some improvement has been obtained.

Results With Two Treatments



Multiple Baseline Design

- a treatment is instituted at successive points in time for two or more persons, settings, or behaviors.
- *Multiple-Baseline-Across-Persons*
 - measure the behavior of each person at baseline
 - measure the behavior of each person at treatment
- *Multiple-Baseline-Across-Settings*
 - measure the behavior in each setting at baseline
 - measure the behavior in each setting at treatment
- *Multiple-Baseline-Across-Behaviors*
 - measure each behavior at baseline
 - measure each behavior at treatment

Benefits & Limitations of Multiple Baseline Designs

- This design is appropriate when the treatment is likely to produce a permanent change in behavior, or when it may be unethical to withdraw the treatment.
- You need more than one person, setting, or behavior to which the treatment can be applied.
- The treatment effect might generalize across the different settings or behaviors occurring before the intervention is instituted.

Criterion-Changing Design

- The effect of the treatment is demonstrated by how closely the behavior matches a criterion that is systematically altered.
- It is appropriate for situations in which the behavior is intended to change gradually.
- It can be greatly strengthened by including periods in which the criterion suddenly changes in the opposite direction.
- Example:
 - Cory decides to gradually reduce his smoking behavior and sets a certain criterion for an allowable number of cigarettes.



MOTIVATING PEOPLE

To Achieve Big Goals, Start with Small Habits

by [Sabina Nawaz](#)

January 20, 2020

Summary

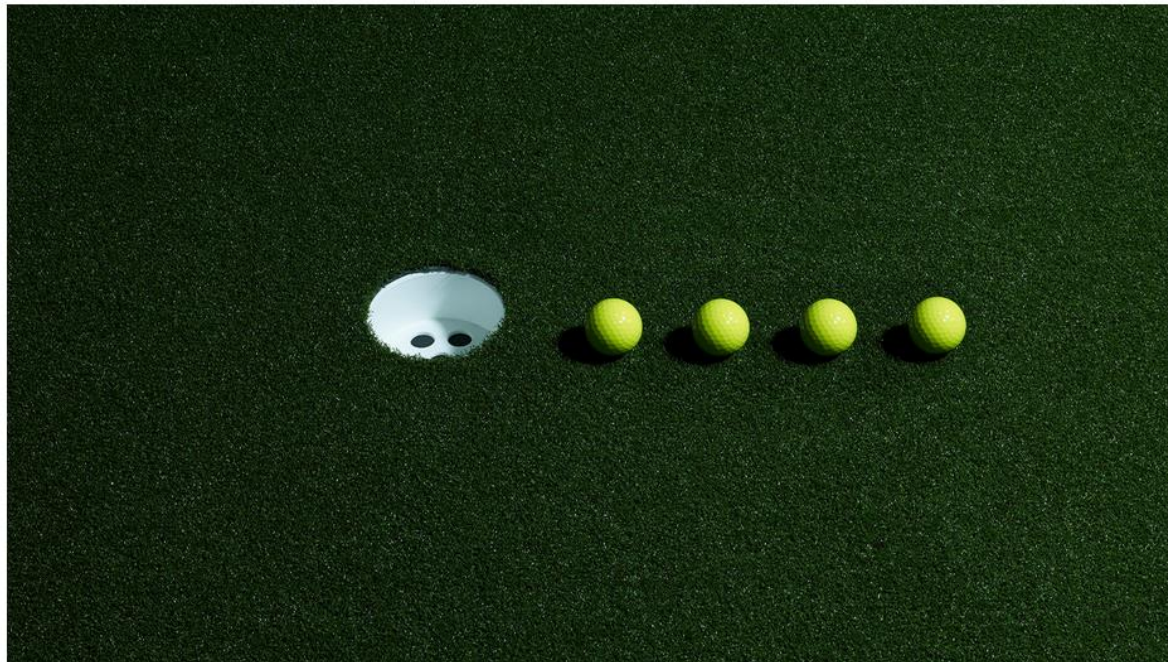
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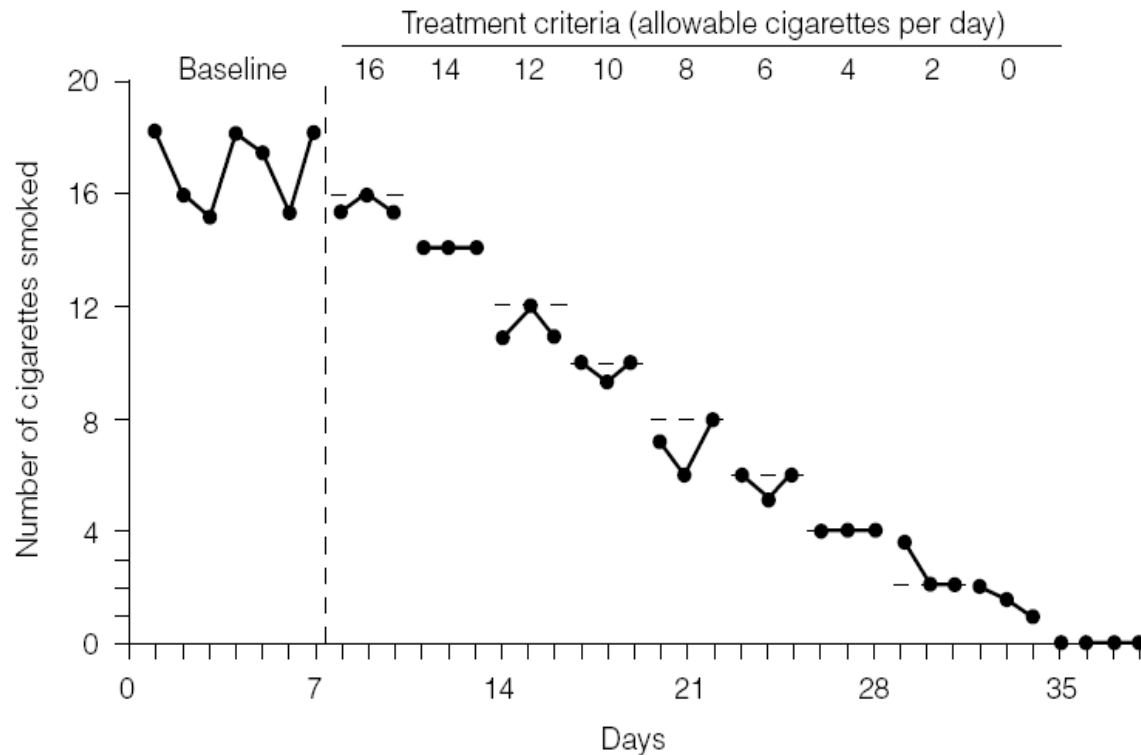
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Cory's Changing Criterion Design Results

FIGURE 2.8 Changing-criterion design. Hypothetical data illustrating use of a changing-criterion design to assess the effectiveness of self-punishment to gradually reduce smoking. The dashed horizontal lines indicate the changing criterion for maximum allowable number of cigarettes smoked per day.



Advantages of Using Animals in Research

- Researchers have the ability to control their genetic makeup.
- Researchers have the ability to control their learning history.
- Researchers have *the ability to control their experimental environment.*
- *Some research cannot ethically be conducted with humans.*

Arguments Against Using Animals in Research

- The findings from animal research necessarily have limited applicability to humans.
- It is morally wrong.
- Animals should have rights similar to humans.